

math111a hw3

jc — October 17, 2025

Problem 1

Find the order of the permutation $(1\ 12\ 8\ 10\ 4)(2\ 13)(5\ 11\ 7)(6\ 9)$

Solution:

(from problem 3) we know that the order must be the LCM of $\{5, 2, 3\}$ which is 30

Problem 2

Let σ be a k -cycle. Prove that σ^i is also a k -cycle if and only if i is relatively prime to k

Solution:

Note that $\sigma^i(a_j) = a_{j+i \bmod k}$ where if $j+i > k$ then we mod $j+i$ by k

This is because, mod k we have

$$\sigma^i(a_j) = \sigma^{i-1}\sigma(a_j) = \sigma^{i-1}(a_{j+1}) = \sigma^{i-2}\sigma(a_{j+1}) = \sigma^{i-2}(a_{j+2}) = \dots = \sigma^{i-i}\sigma(a_{j+i-1}) = a_{j+i}$$

Using this fact, we that continuous applications of σ^i to a_k will yield

$$a_k \mapsto a_{k+i} \mapsto a_{k+2i} \mapsto \dots \mapsto a_{k+(k-1)i} \mapsto a_k$$

Since each $k+xi, k+yi$ are distinct then we have $k+xi \neq k+yi \Rightarrow i(x-y) \not\equiv 0 \bmod k$

Next we need to show that $i(x-y) \not\equiv 0 \bmod k \Leftrightarrow (i, k) = 1$

$\Rightarrow$

Suppose $(i, k) = d > 1$ then let $i = i'd, k = k'd$

Then $i'd(x-y) \not\equiv 0 \bmod k'd \Rightarrow k'd \nmid i'd(x-y) \Rightarrow k' \nmid i'(x-y)$

$\Leftarrow$ is true since $x-y \neq 0$ and i cannot be a factor of k by coprime definition

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Sources:

(1) <https://www.uvm.edu/~mrombach/251-H2-sols.pdf>

(2) <https://math.stackexchange.com/questions/659222/let-sigma-be-the-m-cycle-1-2-ldots-m-prove-that-sigma-i-is-also-a>

Problem 3

Prove the the order of an element in S_n equals the least common multiple of the lengths of the cycle in the cycle decomposition

Solution:

Note that a k cycle will have order k

Therefore, for some k cycle σ we have $\sigma^k = 1 = \sigma^{nk}$ for some $n \in \mathbb{Z}$

If some $\sigma \in S_n$ has (disjoint) cycle decomposition such that $\sigma = \tau_1 \dots \tau_m$ where $\tau_i \in S_n, i \in \{1, \dots, m\}$ then

$$\sigma^r = (\tau_1 \dots \tau_m)^r = \tau_1^r \dots \tau_m^r = 1 \Leftrightarrow \tau_1^r = 1, \dots, \tau_m^r = 1$$

Where the second equality holds due to $\tau_1, \dots, \tau_m$ being disjoint cycles, recall disjoint cycles commute with each other

Therefore the order of σ denoted r must be a multiple of each of $|\tau_1|, \dots, |\tau_m|$

Furthermore, this multiple must be as small as possible, or in other words r is the least common multiple of the orders, or in other words the least common multiple of the lengths of the cycle

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Problem 4

Prove that every $\sigma \in S_n$ can be written as a product of adjacent elements, that is, transpositions of the form $(i, i + 1)$. This was one of the exercises I had given in class

Solution:

From class, we know that every $\sigma \in S_n$ can be written as a product of disjoint cycles

Also, we know that any cycle $(a_1 \ a_2 \ \dots \ a_k) = (a_1 a_2)(a_2 a_3) \dots (a_{k-1} a_k)$ where $a_1 \dots a_k$ are distinct

Any transposition $(a \ b)$ where $a < b$ can be written as (i)

$$(ab) = (a \ a+1) \cdots (b-2 \ b-1)(b-1 \ b)(b-2 \ b-1) \cdots (a \ a+1)$$

In the case of $(a \ b)$ where $a > b$ consider (constructed by the above),

$$(b \ a) = (b, b+1) \cdots (a-2 \ a-1)(a-1 \ a)(a-2 \ a-1) \cdots (b \ b+1)$$

And

$$(b \ a)^{-1} = (b \ b+1) \cdots (a-2 \ a-1)(a-1 \ a)(a-2 \ a-1) \cdots (b \ b+1) = (a \ b)$$

Since $(\sigma_1 \dots \sigma_n)^{-1} = \sigma_n^{-1} \dots \sigma_1^{-1}$ and a two cycle is its own inverse

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example:

$$(1 \ 2)(2 \ 3)(1 \ 2) = (1 \ 3)$$

$$(3 \ 1) = (1 \ 2)(2 \ 3)(1 \ 2)$$

references:

(i) <https://www.albany.edu/~am815139/327.2015.ReviewExam2Sol.pdf>

Problem 5

Prove that for all $n \geq 2$ the set S_n is generated by the elements $(1\ 2)$ and $(1\ 2\ \dots\ n)$

Solution:

Consider that

$$\begin{aligned}(1\ 2\ \dots\ n)(1\ 2)(1\ 2\ \dots\ n)^{-1} &= (2\ 3) \\ (1\ 2\ \dots\ n)(2\ 3)(1\ 2\ \dots\ n)^{-1} &= (3\ 4) \\ &\vdots \\ (1\ 2\ \dots\ n)(n-2\ n-1)(1\ 2\ \dots\ n)^{-1} &= (i\ i+1)\end{aligned}$$

for $1 \leq i \leq n-1$

By problem 4, we know that each $\sigma \in S_n$ can be written as a product of elements of the form $(i\ i+1)$

Since $(1\ 2)$ and $(1\ 2\ \dots\ n)$ generates such elements, they also generate S_n

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Problem 6

Let G be an Abelian group. Prove that the set $\{g \in G : |g| < \infty\}$ is a subgroup of G . This is called the torsion subgroup of G . Give an example where this set is not a subgroup when G is non-abelian

Solution:

Denote $H := \{g \in G : |g| < \infty\}$

We can try to show that $H \neq \emptyset$ and that $\forall x, y \in H$ that $xy^{-1} \in H$

First, note that $1 \in G$ has order 1 so that $H \neq \emptyset$

Also, if $x \in H, y \in H$ let their orders be $|x| = n, |y| = m \Rightarrow |y^{-1}| = m$ then $(xy^{-1})^{nm} = x^{nm}(y^{-1})^{nm} = 1^m 1^n = 1$ and therefore xy^{-1} has order nm and is in H

We can recall that since G is abelian and $x, y \in G$ that $(xy)^n = x^n y^n$ for all $n \in \mathbb{Z}$ and that an element and its inverse have the same order (homework 2) ■

Consider the non-abelian group $GL_2(\mathbb{Q}) = \{A_{2 \times 2}(\mathbb{Q}) : \det A \neq 0\}$

Note that $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has order 4 and $\begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$ has order 3, but $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ which has infinite order (since $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$), and therefore the set $\{g \in GL_2(\mathbb{Q}) : |g| < \infty\}$ is not closed, and therefore not a subgroup

References:

https://en.wikipedia.org/wiki/General_linear_group

Problem 7

- Prove that if H and K are subgroups of a group G , then so is their intersection $H \cap K$
- Prove that the intersection of an arbitrary non-empty collection of subgroups of a group G is again a subgroup of G (do not assume the collection is countable)

Solution:

a) Firstly, $1 \in G \Rightarrow 1 \in H \wedge 1 \in K \Rightarrow 1 \in H \cap K$

Secondly, $a, b \in H \cap K \Rightarrow a \in H \wedge a \in K \wedge b \in H \wedge b \in K \Rightarrow ab \in H \wedge ab \in K \Rightarrow ab \in H \cap K$

Thirdly, $a \in H \cap K \Rightarrow a \in H \wedge a \in K \Rightarrow a^{-1} \in H \wedge a^{-1} \in K \Rightarrow a^{-1} \in H \cap K$

Where each part is justified by properties of subgroups ■

b) Let $H_1 \dots H_2 \dots$ be a collection of subgroups of G , then since each pair of H_i is a subgroup from part a) we know that their intersection $((H_1 \cap H_2) \cap H_3) \cap \dots$ will also be a subgroup, similar to generalized associative law. ■

Problem 8

Let $H_1 \leq H_2 \leq \dots$ be an ascending chain of subgroups of a group G . Prove that $\bigcup_{i=1}^{\infty} H_i$ is a subgroup of G

Solution:

We can combine the following facts:

- a) if $A \subseteq B$ and $B \subseteq C$ then $A \cup B \subseteq C$
- b) $H_1 \cup H_2$ is a subgroup iff $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$

For fact a note that $x \in A \cup B \Rightarrow x \in A \vee x \in B$. If $x \in B \Rightarrow x \in C$ and if $x \in A \Rightarrow x \in B \Rightarrow x \in C$ by the initial assumption

For fact b we can use a weaker version which is that $H_1 \subseteq H_2 \Rightarrow H_1 \cup H_2$ is a subgroup

To show this, consider that (by definition) $H_1 \subseteq H_2 \Rightarrow H_1 \cup H_2 = H_2$ and that H_2 is a subgroup by assumption

Next, $H_1 \cup H_2 \subseteq H_3 \Rightarrow H_1 \cup H_2 \cup H_3 = H_3$ is a subgroup and we can keep applying fact b then fact a and conclude that $\bigcup_{i=1}^{\infty} H_i$ is a subgroup of G

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references:

<https://math.stackexchange.com/questions/3634918/for-two-subgroups-h-1-h-2-how-can-i-show-h-1-cup-h-2-is-subgroup-iff>

Problem 9

In both parts (i) and (ii) prove that the specified subset is not a subgroup of the given group

- (i) the set of 2-cycles in S_n for $n \geq 3$
- (ii) the set of reflections in D_{2n} for $n \geq 3$

Solution:

(i) $(1\ 2)(1\ 3) = (1\ 3\ 2)$ which is a 3 cycle, and the set is not closed

(ii) From homework 2 we know that two successive reflections are equivalent to a rotation $\Rightarrow$ set is not closed